

A SEWER DRAIN SHIELD For Smart Cities

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Drainage systems are vital for storm water management but often transport pollutants, especially plastics, to rivers and oceans, harming ecosystems. Traditional sewer cleaning methods include manual labour, hydraulic flushing, and chemical cleaners. Instead, this sewer drain shield system, combining ultrasonic sensors, GSM modules, nets, and

BILL OF MATERIALS

Components	Quantity
Arduino Uno board (MOD1)	1
GSM module SIM800L (MOD2)	1
Ultrasonic sensor module (S1)	1
USB cable (for connecting 5V)	1
Plastic waste	As required
Wooden ply	As required
PVC pipe	As required
Net	As required

```

1  #include<SoftwareSerial.h>
2
3
4  // Define pins for ultrasonic sensor
5  const int trigPin = 5;
6  const int echoPin = 6;
7
8
9  // Variables for ultrasonic distance calculation
10 long duration;
11 int distance;
12
13
14 // GSM module related configurations
15 SoftwareSerial gsmSerial(9, 10); // RX, TX for GSM module
16 String phone_number = "+919872436823"; // Replace with the number you want to call
17
18
19 void setup() {
20   // Initialize serial communication with Arduino IDE
21   Serial.begin(9600);
22
23
24   // Initialize serial communication with GSM module
25   gsmSerial.begin(4800);
26
27
28   // Define ultrasonic sensor pins
29   pinMode(trigPin, OUTPUT);
30   pinMode(echoPin, INPUT);
31 }

```

Fig. 2: Code snippet setting the pins and number

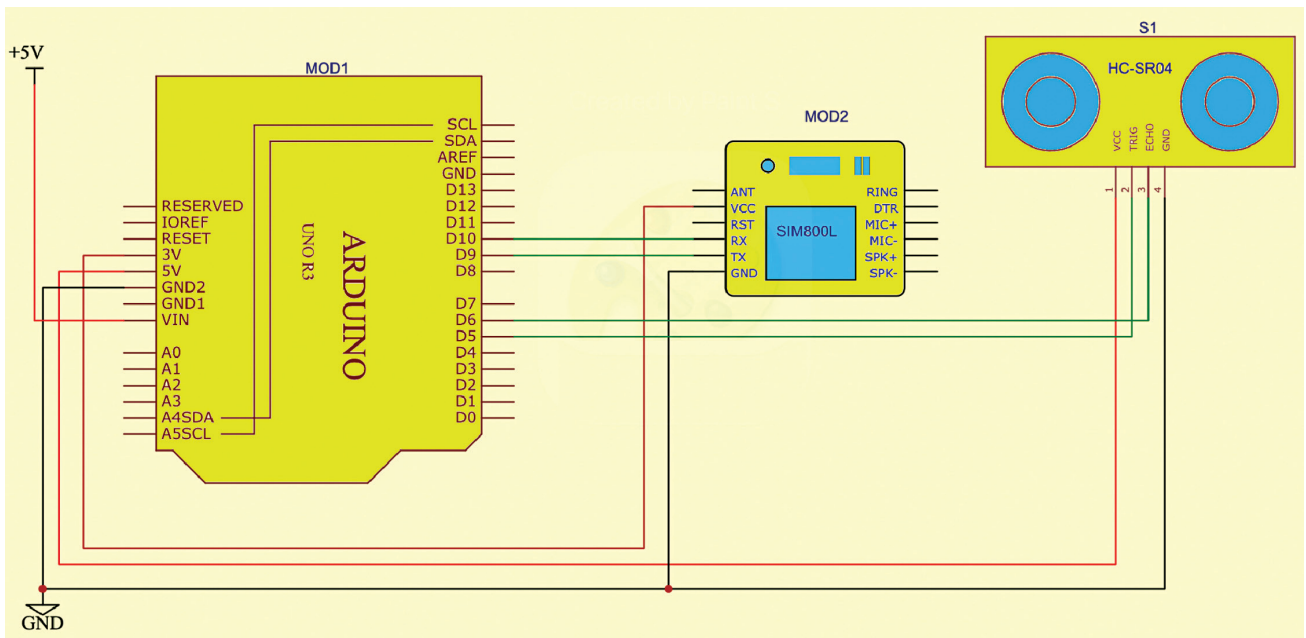


Fig. 1: Circuit diagram